SUHANI

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Portfolio | GitHub | LinkedIn

Professional Summary

Computer Science Engineering student with hands-on experience building backend systems, real-time web applications, and data analytics solutions using Python, JavaScript, React.js, Node.js, SQL, MongoDB, and Scikit-learn. Experienced in developing scalable APIs, analytics dashboards, and network monitoring tools, with additional exposure to machine learning and security vulnerability assessment.

Education

Chandigarh University Chandigarh, India
Bachelor of Engineering in Computer Science Engineering *Expected June 2028*

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence

Technical Skills

Programming Languages: Python, JavaScript, Java, C++, SQL
Frontend Development: React.js, HTML5, CSS3, Monaco Editor API, Responsive Design
Backend Development: Node.js, Express.js, RESTful API Design, Socket.IO, WebSockets
Databases & Analytics: MongoDB, Mongoose, SQL (CTEs, Window Functions, Joins), Power BI, Excel, ETL Pipelines
Machine Learning & Data Science: Scikit-learn, Pandas, NumPy, Random Forest, KNN, Feature Engineering
Networking & Security: Scapy, Socket Programming, TCP/IP, UDP, HTTP, ICMP, Packet Analysis
Developer Tools: Git, GitHub, VS Code, Linux, PyQt, OpenCV, MediaPipe

Projects

E-Commerce Sales Data Analysis & Dashboard | *SQL, Power BI, Python, Pandas, Excel* | GitHub Jan. 2026
- Designed an end-to-end analytics pipeline over 50,000+ transaction records using SQL (CTEs, window functions, joins) to surface revenue trends, seasonal demand patterns, and customer behavior insights.
- Created an interactive Power BI dashboard tracking 8+ KPIs — revenue growth, conversion rate, and churn — used to support data-driven decision-making; standardized data across multiple sources via ETL using Python (Pandas) and Excel.
- Segmented customers using RFM and cohort analysis, identifying high-value groups driving the majority of revenue and enabling more targeted marketing strategies.

Real-Time Collaborative Code Editor | *React.js, Node.js, Express.js, Socket.IO, MongoDB* | GitHub Jan. 2025
- Developed a full-stack collaborative coding platform tested across 10 concurrent rooms, supporting multi-user editing via Socket.IO WebSockets with MongoDB managing persistent session storage.
- Integrated Monaco Editor API with multi-language syntax highlighting and auto-completion; applied debounced state synchronization to keep all connected clients consistent in near real-time.
- Implemented JWT-based authentication, live user presence tracking, and isolated Socket.IO namespaces per room; applied MongoDB indexing that measurably reduced query response time under concurrent load testing.

Network Packet Sniffer & Traffic Analyzer | *Python, Scapy, PyQt, Socket Programming* | GitHub Nov. 2024
- Implemented a network monitoring tool using Python and Scapy to capture and inspect live packet streams across 5 protocol layers — TCP, UDP, HTTP, ICMP, and DNS.
- Engineered a multi-threaded capture engine with configurable protocol filters; added signature-based deep packet inspection (DPI) to flag port scans and suspicious traffic in real time.
- Designed a PyQt GUI displaying live protocol distribution, network statistics, and hex dump logs — reducing manual packet analysis effort during network troubleshooting sessions.

Achievements & Experience

1st Place — CodeMate Hackathon (Data & AI Track): Ranked 1st among 50 teams — trained a Random Forest model with ensemble methods and feature engineering to predict student academic performance, achieving 78% accuracy on a 10,000+ record dataset.

CDAC BOSS Bug Bounty Program: Performed penetration testing and security assessment on government web applications; identified and reported 11 vulnerabilities including XSS and authentication bypass flaws, each documented with step-by-step remediation recommendations.

Certifications & Activities

Certifications: Python for Data Science, AI and Development — IBM (Coursera) | Python Data Analytics — IBM (Coursera) | 5G Network Fundamentals (Coursera)

Activities: Contributed to open-source projects on GitHub including bug fixes, documentation, and feature additions. Active on LeetCode and HackerRank for competitive programming. Multilingual: Fluent in English and Hindi; conversational proficiency in Italian and German.